

Programmer's Model of 8086

Q. Describe the programmer's model of the 8086 microprocessor. Detail the General Purpose Registers, Segment Registers, Pointer/Index Registers, and the Flag Register.

📌 Definition to Remember

The **Programmer's Model** refers to the fourteen 16-bit internal registers accessible to an assembly programmer. They are divided into four groups: **General Purpose**, **Segment**, **Pointer/Index**, and the **Flag Register**.

1. General Purpose Registers (16-bit / 8-bit split)

Can be used as one 16-bit register or two 8-bit registers (High/Low).

- * **AX (Accumulator):** **AH** , **AL** . Heavily used in math (multiplication/division) and I/O.
- * **BX (Base):** **BH** , **BL** . Used to hold base offset addresses for memory access.
- * **CX (Count):** **CH** , **CL** . Used as a loop counter.
- * **DX (Data):** **DH** , **DL** . Used in 32-bit math and holds I/O port addresses.

2. Segment Registers (16-bit)

Hold the base addresses of 64 KB memory segments to help address 1 MB of physical memory.

- * **CS (Code Segment):** Points to program instructions.
- * **DS (Data Segment):** Points to variables/data.
- * **SS (Stack Segment):** Points to the system stack.
- * **ES (Extra Segment):** Additional data segment (used for string destinations).

3. Pointer and Index Registers (16-bit)

Hold offset addresses relative to the segment registers.

- * **SP (Stack Pointer):** Offset of the top of the stack (pairs with SS).
- * **BP (Base Pointer):** Used to pass data via stack (pairs with SS).
- * **SI (Source Index):** Points to source data in string ops (pairs with DS).
- * **DI (Destination Index):** Points to destination data in string ops (pairs with ES).
- * **IP (Instruction Pointer):** Offset of the next instruction (pairs with CS). Not directly modifiable.

4. Flag Register (16-bit)

Contains 9 active flags (6 Status, 3 Control).

- * **Status Flags:**
- * **Carry (CF):** Set if carry out of MSB.
- * **Parity (PF):** Set if lower 8 bits have even number of 1s.
- * **Auxiliary (AF):** Carry from bit 3 to 4 (BCD math).
- * **Zero (ZF):** Set if result is zero.
- * **Sign (SF):** Set if result is negative (MSB=1).
- * **Overflow (OF):** Set if signed overflow occurs.
- * **Control Flags:**
- * **Trap (TF):** Single-step debugging.

* **Interrupt (IF):** Enables maskable interrupts.

* **Direction (DF):** Auto-increment/decrement in string ops.

★ **Must-Write Points (for 10 marks)**

1. Contains **fourteen 16-bit registers**.
2. **General Purpose:** AX (Math/IO), BX (Base address), CX (Loop counter), DX (Data/IO). Can split to 8-bit.
3. **Segment Registers:** CS (Code), DS (Data), SS (Stack), ES (Extra). Base addresses for 64KB segments.
4. **Pointers/Index:** SP & BP (Stack offsets), SI & DI (String offsets), IP (Next instruction offset).
5. **Flag Register:** 16-bit with 9 active flags.
6. **Status Flags:** Carry, Parity, Aux Carry, Zero, Sign, Overflow.
7. **Control Flags:** Trap, Interrupt, Direction.

⚡ **Quick Recall**

Model → General (AX/BX/CX/DX) → Segment (CS/DS/SS/ES) → Pointers (SP/BP/SI/DI/IP) → Flags (6 Status: CF/PF/AF/ZF/SF/OF, 3 Control: TF/IF/DF)